

MUIR AAF, PA, USA

WMO: 725144

Lat: 40.433N

Lon: 76.567W

Elev: 488

StdP: 14.44

Time Zone: -5.00 (NAE)

Period: 07-19

WBAN: 54723

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.2	12.5	-6.5	4.0	11.4	-2.1	5.0	16.3	23.9	32.9	20.5	28.1	6.4	280	0.436

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.7	91.3	73.8	88.7	72.5	86.2	71.5	77.0	85.2	75.5	83.4	74.3	81.7	7.9	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.9	133.1	79.6	73.4	126.3	78.4	72.2	121.3	77.4	40.6	85.1	39.2	83.6	38.0	81.2	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.5	16.9	14.6	DB	1.4	96.3	6.4	3.5	-3.2	98.8	-6.9	100.9	-10.5	102.8	-15.1	105.4	(4)
			WB	0.3	79.9	5.8	1.6	-3.8	81.0	-7.2	81.9	-10.4	82.8	-14.6	84.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.7	28.6	31.7	40.5	52.4	62.2	70.2	74.8	72.3	66.1	54.9	42.8	34.7
	DBStd	17.67	9.47	9.44	9.03	8.51	7.67	5.69	4.87	4.74	6.78	8.08	7.86	7.90
	HDD50	2331	664	517	317	69	4	0	0	0	0	40	241	479
	HDD65	5384	1129	933	759	386	149	20	1	4	68	328	667	940
	CDD50	3322	1	3	24	141	381	606	768	692	484	193	24	5
	CDD65	899	0	0	0	8	61	176	304	231	103	16	0	0
	CDH74	7879	0	1	6	163	612	1580	2844	1763	806	104	0	0
	CDH80	2687	0	0	0	44	175	500	1144	544	258	22	0	0
Wind	WSAvg	5.0	6.0	6.2	6.6	6.6	4.7	4.5	3.8	3.3	3.6	4.5	5.0	5.5
Precipitation	PrecAvg	45.0	3.2	2.6	3.7	3.6	4.0	4.1	4.4	4.3	4.5	4.0	3.0	3.6
	PrecMax	71.9	6.5	5.6	6.8	9.0	6.7	10.0	7.8	8.1	16.3	9.8	5.3	6.7
	PrecMin	29.1	1.4	0.6	0.9	1.1	1.1	0.9	1.5	1.3	0.2	0.7	0.8	0.7
	PrecStd	8.2	1.3	1.3	1.6	1.7	1.5	1.9	1.6	1.7	3.3	2.2	1.2	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.6	65.0	73.3	85.6	89.0	93.2	96.4	91.5	91.1	82.8	69.0	61.6
		MCWB	55.2	53.8	56.6	64.1	71.8	75.4	74.4	73.6	74.0	69.8	58.8	59.3
	2%	DB	49.7	56.4	66.8	78.8	84.8	88.7	92.7	88.4	87.1	76.5	63.5	55.2
		MCWB	45.6	49.2	55.1	62.1	69.7	72.2	74.6	73.0	71.9	66.9	56.7	52.7
	5%	DB	45.6	50.0	60.4	72.7	81.2	85.5	89.4	85.8	82.5	72.3	59.6	50.4
		MCWB	42.7	43.5	51.3	58.7	67.8	70.6	73.2	72.0	69.9	64.4	52.8	47.3
	10%	DB	42.0	44.2	55.1	67.9	76.8	82.7	86.4	83.1	78.6	68.5	55.4	46.4
		MCWB	38.6	39.0	47.5	55.6	65.6	69.1	71.9	70.6	67.9	61.8	49.4	43.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.6	60.1	60.7	69.9	76.1	77.3	79.1	78.3	77.2	72.7	62.3	59.2
		MCDB	57.8	63.9	67.3	76.3	82.7	86.8	90.3	85.4	84.9	77.2	64.3	60.7
	2%	WB	46.9	51.0	57.7	64.9	72.7	75.3	77.1	76.5	74.7	70.3	58.8	53.8
		MCDB	47.7	54.5	65.2	73.2	80.6	84.6	86.4	83.1	81.7	74.4	62.3	54.5
	5%	WB	42.9	44.3	53.1	61.9	70.4	73.6	75.5	74.7	72.7	67.3	54.7	48.2
		MCDB	45.2	48.6	58.7	70.4	77.6	81.5	84.0	81.0	78.9	71.1	58.1	49.3
	10%	WB	38.9	40.0	48.6	58.5	68.3	71.9	74.3	73.1	70.7	63.2	50.9	43.5
		MCDB	41.3	43.3	53.4	66.1	73.9	79.1	82.3	79.1	76.0	66.7	53.9	45.3
Mean Daily Temperature Range	5% DB	MDBR	14.7	16.3	18.0	20.8	20.2	20.6	20.7	19.4	20.2	19.3	16.9	13.5
		MCDBR	18.1	24.1	26.8	29.6	26.8	24.7	24.7	22.6	24.9	23.7	23.6	18.1
		MCWBR	15.9	17.9	16.8	15.3	13.4	10.9	9.6	9.7	11.4	13.8	15.9	16.2
	5% WB	MCDBR	16.8	22.7	23.1	25.2	22.6	21.4	20.5	18.7	19.8	18.8	18.9	16.0
		MCWBR	16.0	18.3	17.0	15.6	13.1	11.1	9.1	9.2	10.7	12.8	15.6	16.2
Clear-Sky Solar Irradiance	taub	0.311	0.332	0.366	0.392	0.425	0.493	0.498	0.492	0.411	0.369	0.328	0.309	
	taud	2.422	2.336	2.328	2.290	2.228	2.073	2.071	2.076	2.275	2.401	2.480	2.471	
	Ebn at Noon	269	278	280	279	272	252	249	247	262	262	261	261	
	Edh at Noon	26	32	36	41	44	52	51	50	38	30	24	23	
All-Sky Solar Radiation	RadAvg	567	853	1145	1482	1669	1845	1872	1629	1333	936	654	477	
	RadStd	48	110	86	142	197	157	102	91	178	106	64	58	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (60 neighbors)		+0.72	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+212	+115

Nomenclature: See separate page